

# Introduction to Python Programming

## Lec - 7

# Objects and Dynamic Typing

- Every object has a type and a value

```
In [1]: type(7)
Out[1]: int

In [2]: type(4.1)
Out[2]: float

In [3]: type('dog')
Out[3]: str
```

- An object's value is the data stored in the object. The snippets above show objects of Python built-in types `int` (for integers), `float` (for floating-point numbers) and `str` (for strings)

# Variables Refer to Objects

- Assigning an object to a variable binds (associates) that variable's name to the object. As you've seen, you can then use the variable in your code to access the object's value.

```
In [4]: x = 7  
  
In [5]: x + 10  
Out[5]: 17  
  
In [6]: x  
Out[6]: 7
```

```
In [7]: x = x + 10  
  
In [8]: x  
Out[8]: 17
```

- After snippet [4]'s assignment, the variable `x` refers to the integer object containing 7. As shown in snippet [6], snippet [5] does not change `x`'s value. You can change `x` as follows

# Intro to Data Science: Basic Descriptive Statistics

- minimum—the smallest value in a collection of values.
- maximum—the largest value in a collection of values.
- range—the range of values from the minimum to the maximum.
- count—the number of values in a collection.
- sum—the total of the values in a collection.

Q : WAP to find min and max of 3 values. Use in built function, and without it.

Q Solve the previous question by taking input as a list.

# Auxiliary Content

- Write a loop to print numbers from 0->10
- Write a loop to print numbers from 10->0
- Write a loop to print even numbers from (m->n)
- Write a loop to print odd numbers from (n->m) in descending order
- Write a loop to print characters of a string
- Write a loop to print characters of a string where we skip the character 'L'.
- Write a loop to print characters of a string where we stop printing until we reach the end of the string or we encounter a 'J'
- Write a loop to print a 2D array where there are x rows and y columns